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$$M(t) = A \cos(\omega_n t) + B \sin(\omega_n t)$$

$$M(t) = A \omega_n \sin(\omega_n t) + B \omega_n \cos(\omega_n t)$$

a.) $M(0) = 0$

$$M(0) = V_0$$

$$\xi = 0$$

Encontramos "A"

$$M(0) = A \cos(0) + B \sin(0)$$

$$0 = A(1) + B(0)$$

$$A = 0$$

Encontramos "B"

$$M(0) = A \omega_n \sin(0) + B \omega_n \cos(0)$$

$$V_0 = B \omega_n$$

$$B = V_0 / \omega_n$$

$$M(t) = \frac{V_0}{\omega_n} \sin(\omega_n t) \quad \text{Ecuación de movimiento}$$

b.)

$$u(0) = u_0 + B \sin(\omega_n t)$$

$$u(0) = v_0 + B \omega_n \cos(\omega_n t)$$

Encontramos "A"

$$u_0 = A \cos(0) + B \sin(0)$$

$$u_0 = A$$

$$u_0 = \omega_0$$

Encontramos B

$$v_0 = -A \omega_n \sin(0) + B \omega_n \cos(0)$$

$$v_0 = B \omega_n$$

$$B = \frac{v_0}{\omega_n}$$

$$u(t) = u_0 \cos(\omega_n t) + \frac{v_0}{\omega_n} \sin(\omega_n t)$$

Función de Movimiento